SUPPOSITIONS

STANDARD PROOF TECHNIQUES IN DISCRETE MATHEMATICS AND EXAMPLE APPLICATIONS

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ABSTRACT

This document examines standard proof techniques found in discrete mathematics. It combines meticulous explanations with example applications to accessibly define and demonstrate each approach.

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1 EXISTENTIAL STATEMENTS

1.1 PROOF BY CONSTRUCTION

2 UNIVERSAL STATEMENTS

2.1 PROOF BY EXHAUSTION

2.2 DIRECT PROOF

2.3 PROOF BY DIVISION INTO CASES

2.4 PROOF BY CONTRAPOSITION

2.5 PROOF BY CONTRADICTION

2.6 PROOF BY MATHEMATICAL INDUCTION

2.7 PROOF OF SUBSET CONTAINMENT

2.8 PROOF OF INJECTIVITY

2.9 PROOF OF SURJECTIVITY

2.10 PROOF OF BIJECTIVITY

3 EXAMPLE PROBLEMS

3.1 SQUARES OF ODD INTEGERS

Exercise. Prove that if a is an odd integer, then $a^2 = 8m + 1$ for some $m \in \mathbb{Z}$.

Proof. This statement can be proven by division into cases.

SETUP.

Let a be an odd integer. Then $\exists k \in \mathbb{Z}$ such that $a = 2k + 1$. Thus,

$$\begin{aligned} a^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 4k(k+1) + 1. \end{aligned}$$

Either k is even or odd.

CASE 1: k IS EVEN.

Then $\exists n \in \mathbb{Z}$ such that $k = 2n$. Thus,

$$\begin{aligned} a^2 &= 4k(k+1) + 1 \\ &= 4(2n)[(2n)+1] + 1 \\ &= 4(4n^2 + 2n) + 1 \\ &= 8(2n^2 + n) + 1. \end{aligned}$$

Let $m = 2n^2 + n$. Then $m \in \mathbb{Z}$ and $a^2 = 8m + 1$.

CASE 2: k IS ODD.

Then $\exists n \in \mathbb{Z}$ such that $k = 2n + 1$. Thus,

$$\begin{aligned} a^2 &= 4k(k+1) + 1 \\ &= 4(2n+1)[(2n+1)+1] + 1 \\ &= 4(2n+1)(2n+2) + 1 \\ &= 4(4n^2 + 6n + 2) + 1 \\ &= 8(2n^2 + 3n + 1) + 1. \end{aligned}$$

Let $m = 2n^2 + 3n + 1$. Then $m \in \mathbb{Z}$ and $a^2 = 8m + 1$.

In either case, $a^2 = 8m + 1$ for some $m \in \mathbb{Z}$. Therefore, the statement holds. $\square$

3.2 DISTANCES FROM A FIXED POINT

Exercise. Prove that if $x < -4$ and $y > 2$, then the distance from (x, y) to $(1, -2)$ is at least 6.

Proof. This statement can be proven directly.

SETUP.

Let $x, y \in \mathbb{R}$ such that $x < -4$ and $y > 2$. Let d denote the distance from (x, y) to $(1, -2)$. By the Euclidean distance formula,

$$d = \sqrt{(1-x)^2 + (-2-y)^2}.$$

BOUND 1: WE KNOW $x < -4$.

Then

$$\begin{aligned} -x &> 4 \\ -x + 1 &> 4 + 1 \\ 1 - x &> 5 \\ (1-x)^2 &> 5^2 \\ (1-x)^2 &> 25. \end{aligned}$$

BOUND 2: WE KNOW $y > 2$.

Then

$$\begin{aligned} -y &< -2 \\ -y - 2 &< -2 - 2 \\ -2 - y &< -4 \\ (-2-y)^2 &> (-4)^2 \\ (-2-y)^2 &> 16. \end{aligned}$$

COMBINING THE BOUNDS.

By the first and second bounds,

$$\begin{aligned} d &= \sqrt{(1-x)^2 + (-2-y)^2} \\ &> \sqrt{25+16} \\ &= \sqrt{41} \\ &\geq 6. \end{aligned}$$

Therefore, the distance from (x, y) to $(1, -2)$ is at least 6. □

3.3 LATTICE POINTS IN CIRCLES

Exercise. A point $(m, n) \in \mathbb{R}^2$ is a lattice point if $m, n \in \mathbb{Z}$. Prove that the number of lattice points inside any circle centered at the origin equals $4k + 1$ for some $k \in \mathbb{Z}$.

Proof. Hello! □

3.4 DOT PRODUCTS OF FACTORIALS

Exercise. For $(a_1, a_2, a_3, \ldots, a_n), (b_1, b_2, b_3, \ldots, b_n) \in \mathbb{R}^n$, the dot product is

$$(a_1, a_2, a_3, \ldots, a_n) \cdot (b_1, b_2, b_3, \ldots, b_n) = a_1b_1 + a_2b_2 + a_3b_3 + \cdots + a_nb_n.$$

For example,

$$(5, 2, -3) \cdot (3, 7, 4) = 5 \cdot 3 + 2 \cdot 7 + (-3) \cdot 4 = 17.$$

Prove that

$$(1, 2, 3, \ldots, n) \cdot (1!, 2!, 3!, \ldots, n!) = (n + 1)! - 1$$

for all positive integers n.

Proof. This statement can be proven by mathematical induction.

BASE CASE: LET $n = 1$.

Then

$$\begin{aligned}(1, 2, 3, \ldots, n) \cdot (1!, 2!, 3!, \ldots, n!) &= (1) \cdot (1!) \\ &= 1 \cdot 1 \\ &= 1,\end{aligned}$$

and

$$\begin{aligned}(n + 1)! - 1 &= (1 + 1)! - 1 \\ &= 2! - 1 \\ &= 2 - 1 \\ &= 1.\end{aligned}$$

Thus,

$$(1, 2, 3, \ldots, n) \cdot (1!, 2!, 3!, \ldots, n!) = (n + 1)! - 1.$$

INDUCTIVE HYPOTHESIS.

$\exists k \in \mathbb{Z}$ with $k \geq 1$ such that

$$(1, 2, 3, \ldots, k) \cdot (1!, 2!, 3!, \ldots, k!) = (k + 1)! - 1.$$

INDUCTION: CONSIDER $k+1$.

We want to demonstrate that

$$[1, 2, 3, \ldots, k, (k+1)] \cdot [1!, 2!, 3!, \ldots, k!, (k+1)!] = [(k+1)+1]! - 1.$$

Per the inductive hypothesis and sequence definition,

$$\begin{aligned}
[1, 2, 3, \ldots, k, (k+1)] \cdot [1!, 2!, 3!, \ldots, k!, (k+1)!] &= [(k+1)! - 1] + (k+1)(k+1)! \\
&= (k+1)! + (k+1)(k+1)! - 1 \\
&= (k+1)! \cdot [1 + (k+1)] - 1 \\
&= (k+1)! \cdot (k+2) - 1 \\
&= (k+2)! - 1 \\
&= [(k+1)+1]! - 1.
\end{aligned}$$

Therefore, the statement holds by the principle of mathematical induction. □

3.5 LEFT INVERSES OF INJECTIONS

Exercise. Let A be a nonempty set. Prove that a function $f\colon A \to B$ is injective if and only if f has a left inverse, that is, a function $g\colon B \to A$ such that $(g \circ f)(x) = x$ for all $x \in A$.

Proof. Hello! □

3.6 INTERSECTIONS OF OPEN SETS

Exercise. A set $S \subseteq \mathbb{R}$ is open if for each $b \in S$, there exists $\epsilon > 0$ such that $(b-\epsilon, b+\epsilon) \subseteq S$. Prove that the intersection of two open sets is open.

Proof. Hello! □